THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET A

Chapter 7: **Particulate Nature of Matter** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** extremely small particles	Particulate nature
2	**(d)** gases	Negligible forces
3	**(a)** no fixed shape and no fixed volume	Particles fully free
4	**(c)** large interparticle spaces exist	Particles come closer
5	**(d)** 0 °C	As in the textbook table
6	**(c)** vibrate more vigorously	Energy gain
7	**(a)** spread among air particles	Gases mix easily
8	**(d)** its particles are mixed in very small sizes among water particles	Particles dissolve and spread
9	**(d)** a solid made of tiny solid grains	Each grain has a fixed shape
10	**(d)** liquid	Solid melts and can then vaporise

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Matter is made of extremely small particles (1). There are spaces between them and forces between them (1).	1+1
12	Gaps between particles (1). Solid state (1).	1+1
13	Strong interparticle forces (1). Fixed shape and volume (1).	1+1
14	Liquid particles can move past each other (1). Solid particles are held in fixed positions (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Solids: fixed shape and volume, least space (1). Liquids: fixed volume, no shape (1). Gases: neither, most space (1).	1+1+1
16	Solids: vibrate about fixed positions (1). Liquids: move within a space (1). Gases: move freely in all directions (1).	1+1+1
17	Ice: regular, close (1). Water: loosely arranged, mobile (1). Vapour: far apart and free (1).	1+1+1
	OR: On heating, particles vibrate more (1). Forces weaken and particles leave fixed positions (1). Temperature at which this occurs is melting point (1).	1+1+1
18	Particles move vigorously on heating (1). Interparticle forces decrease (1). Particles escape as vapour at boiling point (1).	1+1+1
	OR: Fill syringe with air, push plunger (1). Air compresses; particles come closer (1). Water cannot be compressed because it has little space (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Matter is made of small particles (1). Solid arrangement and movement (1). Liquid arrangement and movement (1). Gas arrangement and movement (1). Forces strongest to weakest (1).	1+1+1+1+1
	OR: Shape (1). Volume (1). Compressibility (1). Flow (1). Interparticle space with forces (1).	1+1+1+1+1
20	Melting: vibrations increase (1). Forces weaken (1). Melting point definition (1). Boiling: particles escape as vapour (1). Boiling point definition (1).	1+1+1+1+1
	OR: Syringe with air compresses (2). Syringe with water does not (1). Solid blocks cannot be compressed (1). Explain by interparticle space (1).	2+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET B

Chapter 7: **Particulate Nature of Matter** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** interparticle spaces	Gaps between particles
2	**(c)** gases	Particles far apart
3	**(d)** vibrate about fixed positions	Fixed positions
4	**(a)** come closer	Space decreases
5	**(b)** 1538 °C	Strong particle attraction
6	**(d)** become weaker	Particles leave fixed positions
7	**(c)** particles of ink and water mix	Particles move and intermingle
8	**(c)** Its particles are farther apart than those in water	Ice is an exception
9	**(d)** nothing of the chair remains	Matter is made of particles
10	**(b)** its particles move freely in all directions	Negligible attraction

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Interparticle forces are negligible (1). Particles move freely and fill the container (1).	1+1
12	Temperature at which a solid melts at atmospheric pressure (1). 0 °C (1).	1+1
13	Temperature at which a liquid boils (1). It changes to vapour at atmospheric pressure (1).	1+1
14	Gas particles are far apart (1). Water particles are close with little space (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Gas particles move freely and have large spaces (1½). Solid particles are tightly held (1½).	1½+1½
16	Strong interparticle attraction (1). Minimum space and no free movement of particles (2).	1+2
17	Liquid particles are close, so volume is fixed (1½). Particles move past each other so shape changes (1½).	1½+1½
	OR: No, they are solids (1). Each grain or particle of flour keeps its own shape (1). They only pile in the container (1).	1+1+1
18	Aluminium: closely packed regular (1). Glycerin: close but irregular (1). Methane: far apart (1).	1+1+1
	OR: Salt breaks into tiny particles in water (1½). They spread throughout water and are not visible (1½).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Ink in water spreads (1½). Perfume spreads in a room (1½). Sugar dissolves and sweetens water (1). Particles move on their own (1).	1½+1½+1+1
	OR: Ice melts at low temperature (1). Urea at medium (1). Iron very high (1). High melting point means strong forces (1). Energy needed is more (1).	1+1+1+1+1
20	Solid forces strong (1). Liquids moderate (1). Gases negligible (1). Spaces small, small, large (1). Result: rigid, flow, spread (1).	1+1+1+1+1
	OR: Ice – close and regular (1). Water – close and mobile (1). Vapour – far apart (1). Melting and freezing (1). Boiling and condensation (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.